What BookTrap costs to run

Monthly spend at 100, 1,000 and 10,000 readers, from what Phase 9 measured: your real reading costs, a load test at Render's limits, and the storage one shelf takes.

$54
a month at launch: 100 readers pasting 5 posts each ($29 on Supabase's free plan)

$202
a month at 1,000 readers, 5 posts each

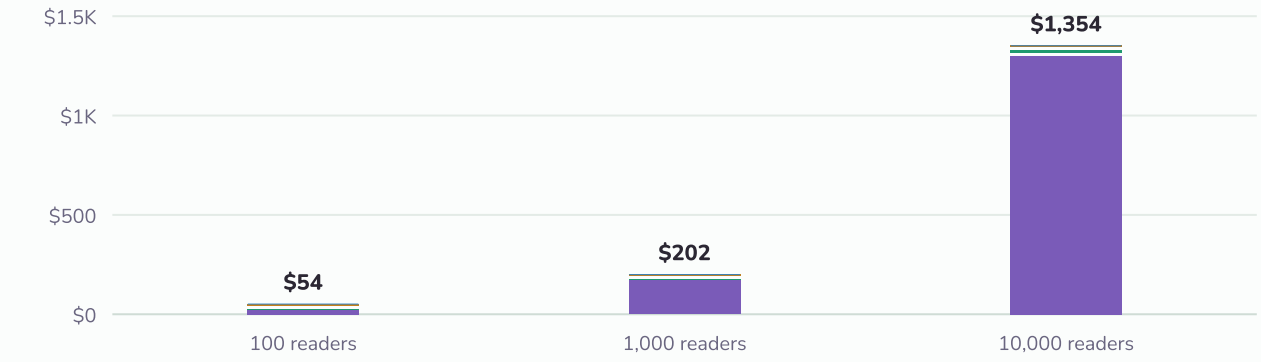
$1,354
a month at 10,000 readers, 5 posts each

The short version

- **Reading posts is almost all of it.** Each new post costs about $0.05 (Claude reading it, Apify fetching it). At 1,000 readers that's 84% of the bill. Hosting and the database stay small.
- **Your $20 reading budget covers about 384 new posts a month.** 100 readers at 5 posts each need $21, so the budget is about right at launch and reading pauses for everyone past it.
- **Hosting fits the $10 target up to about 1,000 readers** on Render Starter ($7). It reads about 174 posts an hour; past that, Standard ($25) doubles it.
- **Testing has cost $0.78 in total** since 8 October, all real reads during Phases 2 to 7. The load test and Phases 8 and 9 cost nothing.

Where the money goes

One month at each size, 5 posts per reader.



Readers	Reading posts	API hosting	Database	Email and domain	Total	Per reader
100	$21	$7.00	$25	$1.00	**$54**	$0.54
1,000	$169	$7.00	$25	$1.00	**$202**	$0.20
10,000	$1,300	$28	$25	$1.00	**$1,354**	$0.14

If readers paste more, or less

Posts per reader a month	100 readers	1,000 readers	10,000 readers
2	$41 (reading $8.32, Starter)	$101 (reading $68, Starter)	$556 (reading $520, Starter)
5	$54 (reading $21, Starter)	$202 (reading $169, Starter)	$1,354 (reading $1,300, Standard)
10	$75 (reading $42, Starter)	$371 (reading $338, Starter)	$2,714 (reading $2,600, Pro)
20	$116 (reading $83, Starter)	$709 (reading $676, Starter)	$5,314 (reading $5,200, Pro)

With Supabase Pro ($25: daily backups, never paused). Render plan chosen so the busiest hour uses under 70% of what it can read.

Ways to spend less per post

- A smaller Claude model for posts whose caption lists the books; Sonnet for videos.
- 20 frames per video instead of 30: about a third less image input.
- Charge for heavy use: the planned paid option for big collections, or a plan above the free posts.

What I'd set at launch

- Render Starter ($7), reading 3 posts at a time.
- Supabase Pro ($25) once real readers arrive, for backups. Free works for testing.
- Keep the $20 reading budget and watch the spend page; raise it once readers stay.

How many readers one server handles

100 readers opened BookTrap within the same 10 seconds and each pasted a video. The API ran in Docker at Render's limits against the real database. Apify and Claude were stubbed, so nothing was paid for, but every video went through the real frame reader (ffmpeg on a 60-second 720p video).

Render plan	Price	At once	Posts read	Last of the batch read after	Peak memory	Errors
Starter (0.5 CPU, 512 MB)	$7	2	100/100	32 min	235 MB	0
Starter, lower-priority ffmpeg	$7	3	100/100	35 min	297 MB	0
Starter (40 readers)	$7	4	40/40	13 min	326 MB	0
Standard (1 CPU, 2 GB)	$25	2	99/100	16 min	304 MB	1

The one error on Standard: a single paste out of 100 waited 45 seconds and failed with a server error during the opening burst. Its log was lost when the next run started, so the cause isn't known; the harness now keeps every run's log, and no later run repeated it.

- **The CPU is the limit, not memory.** Starter reads about 174 video posts an hour however many it reads at once; 4 at once used 326 MB of 512. Standard read twice as fast.

- **Most of a read is ffmpeg decoding the video** (about 10 of 14 seconds at 0.5 CPU). Decoding only keyframes would be 70× faster but gives fewer frames and may miss books, so it isn't used.
- **The slow requests were the arrival burst, not the reading.** When all 100 readers opened BookTrap in the same 10 seconds, a paste took 4.1s (worst 16.4s). Once running, a progress check answered in 0.45s even with ffmpeg busy. Running ffmpeg at a lower priority made no measurable difference (it stays, as it costs nothing). Each request makes a few trips to the database, here from Bahrain to Singapore; on Render, next to the database, they're much shorter.
- No temp files were left behind, and no post was lost.
- **Cheap fix to make:** the API doesn't compress its responses yet. A 1,000-book shelf is 1,106 KB as sent; compressed it would be several times smaller, which keeps traffic inside Render's free 5 GB for longer.

Storage

One book on a shelf, with its sources, shelves and indexes, takes about 1.62 KB (measured with a 500-book shelf). A reader with 150 books and a month of history is about 0.24 MB. Opening a 1,000-book shelf sends 1106 KB and took 0.8s.

Readers	Database size	Fits in
100	39 MB	Supabase Free (500 MB)
1,000	266 MB	Supabase Free (500 MB)
10,000	2.5 GB	Supabase Pro (8 GB included)
100,000	24.7 GB	Pro plus 17 GB ($2.13 a month)

What testing has cost

$0.78
all paid reading since 8 October: Claude $0.52 (11 reads, $0.047 each), Apify $0.26

$0
Phases 8 and 9: cached posts and stubbed services only

$0
hosting: Render offline; Supabase and Cloudflare free; the load test ran in Docker on this Mac

From BookTrap's own spend records, which log every paid Apify run and Claude read. The Phase 2 model comparison ran outside the app and isn't in them. Your Claude subscription isn't included.

Assumptions

Assumption	Value	Source
Reading one new post	$0.05	Claude $0.047 (average of your 11 reads) + Apify about $0.006 (TikTok) or $0.003 (Instagram)
Posts already read by someone (free)	20%, 35%, 50%	Estimate for 100, 1,000 and 10,000 readers: popular posts get pasted again
Busiest hour	12% of a day	Estimate; picks the Render plan
Posts read an hour	Starter 174, Standard 360, Pro 720	Starter and Standard measured; Pro estimated as twice Standard
Traffic per reader a month	2 MB	Opening a 150-book shelf about 12 times, plus following reads. Render's free workspace includes 5 GB, then $0.15/GB
Sign-in emails per reader	0.3 a month	A code is only sent on a new browser. Resend is free up to 3,000 a month, then $20
Domain	$1 a month	About $12 a year

Prices checked on 11 October 2026: Render (Starter $7, Standard $25, Pro $85; a free workspace includes 5 GB of outbound traffic), Supabase (Free: 500 MB, paused after a week idle; Pro $25 with 8 GB, then $0.125/GB), Resend (free to 3,000 emails a month).

numbers will firm up after launch